

PROBLEM STATEMENT

The Missionaries and Cannibals Problem is a classic Artificial Intelligence problem in which **three missionaries** and **three cannibals** are standing on one side of a river and must cross to the other side using a boat. The boat can carry a maximum of **two people** at a time and cannot cross the river without at least one person on board.

The objective is to transport all the missionaries and cannibals safely to the opposite side while satisfying the following constraints:

- The boat can carry either one or two people.
- At no point on either river bank should the number of cannibals exceed the number of missionaries, unless there are no missionaries on that bank.
- The boat must travel back and forth until everyone reaches the destination safely.

The goal is to determine a valid sequence of crossings that satisfies all the constraints.